Weekly Schedule

Spring 2027 · Mondays are lecture, Wednesdays are tutorials

> **Note**
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> **Mondays:** lecture (slides linked below). **Wednesdays:** hands-on tutorial in class, which feeds directly into that week's assignment. Data Safari exercises appear in Weeks 4, 7, 10, and 13, alongside the regular assignment.
>
> Each assignment/Data Safari link (e.g. "PS1") downloads the blank Quarto template (`.qmd`) — fill it in and render it yourself to produce your submission PDF. Tutorials and slides can be viewed with **Code → View Source** (top right of the page) if you'd like their `.qmd` source too.

Week	Topic	Slides	Tutorial	Assignment	Data Safari
1	Why experimental design matters; setting up R, RStudio, and Quarto	Slides	Tutorial	PS1	
2	R fundamentals and data wrangling	Slides	Tutorial	PS2	
3	Exploring data I: distributions and summary statistics	Slides	Tutorial	PS3	
4	Exploring data II: relationships and group comparisons	Slides	Tutorial	PS4	Safari 1
5	Principles of experimental design: repli-	Slides	Tutorial	PS5	

Week	Topic	Slides	Tutorial	Assignment	Data Safari
	cation, randomization, blocking				
6	Design in practice: sketching designs from the published literature	Slides	Tutorial	PS6	
7	Hypothesis testing I: the t-test and one-way ANOVA	Slides	Tutorial	PS7	Safari 2
8	**Spring Break**				
9	Correlation and simple linear regression	Slides	Tutorial	PS9	
10	Multiple regression and ANCOVA	Slides	Tutorial	PS10	Safari 3
11	Generalized linear models: logistic regression for binary/ count data	Slides	Tutorial	PS11	
12	Accounting for non-independence: introduction to mixed-effects models	Slides	Tutorial	PS12	
13	Mixed-effects models II: ran-	Slides	Tutorial	PS13	Safari 4

Week	Topic	Slides	Tutorial	Assignment	Data Safari
	dom slopes, nested vs. crossed designs				
14	Nonparametric alternatives and choosing the right test	Slides	Tutorial	PS14	
15	**Final Project Presentations**				

Graduate project milestones (BIOL 5860 only)

Four checkpoint deliverables spread across the semester, each documenting one stage of your independent project. See the Assignments overview for the full description. As with weekly assignments, click the name to download the blank `.qmd` template.

Due by end of	Milestone	Template
Week 7	Experimental design and planned hypothesis testing	Part 1
Week 10	Data carpentry and exploration	Part 2
Week 13	Statistical modeling	Part 3
Week 14	Visualization	Part 4
Week 15	Final presentation covering the entire process	—

> **Note**
>
> These due dates are a suggested pacing (each milestone lands a week after the most relevant course content), not fixed by any particular constraint — adjust freely in this table and in the corresponding files' front matter if you'd prefer a different spread.

Notes

- Slides open as a separate reveal.js presentation (use arrow keys or click to advance); tutorials and assignments render as regular pages on this site.

- The graduate-only Project is presented in Week 15, alongside undergraduate project presentations.